# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 300 | 1 crore = 100 lakh. |
| 2 | **(c)** 18 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 3 | **(d)** 1 | Multiply as whole numbers, then count the decimal places in both factors. |
| 4 | **(b)** 28 | Perimeter $= 4\times$ side. |
| 5 | **(a)** 2 | Group digits in threes. |
| 6 | **(c)** 23 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 7 | **(b)** 8 | Multiply both numbers by 10 or 100 so that the divisor becomes a whole number. |
| 8 | **(a)** $5n + 2$ | Write the rule using a letter-number. |
| 9 | **(a)** 4,560,000 | Regroup the digits in threes. |
| 10 | **(d)** 7 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 11 | **(b)** 345 | 1 m = 100 cm. |
| 12 | **(b)** 16 | Combine the like terms first: $m + am = (1 + a)m$. |
| 13 | **(b)** 4,00,000 | Digit times its place value. |
| 14 | **(b)** $(3 + 4) \times 5 - 6$ | Check which grouping gives the value. |
| 15 | **(d)** 2500 | 1 kg = 1000 g. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Regroup in threes (1); 2,345,678 (1). | 1+1 |
| 17 | Evaluate inner brackets first; Then the outer operations (1); Value $= 32$ (1). | 1+1 |
| 18 | Multiply without the decimal points: $32 \times 5 = 160$ (1); Decimal places in the factors: $1 + 1 = 2$, so the product is $1.6$ (1). | 1+1 |
| 19 | Substitute $n$ in $3n + 1$ (1); $3(5) + 1 = 16$ matchsticks (1). | 1+1 |
| 20 | Successor of 99,999 is 1,00,000 (1); Successor of 99,999 is 1,00,000 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Break the number by place value (1); $3 \times 10^6 + 4 \times 10^5 + 5 \times 10^4 + 6 \times 10^3 + 7 \times 10^2 + 8 \times 10^1 + 9 \times 10^0$ (1); In lakhs: $34.6$ lakh (1). | 1+1+1 |
| 22 | Innermost bracket first; Then outward (1); Then multiply/divide, add/subtract (1); Value $= 28$ (1). | 1+1+1 |
| 23 | Cost $= 125.5 \times 3.5$ (1); Cost $=$ ₹$439.25$ (2). | 1+2 |
| 24 | Figure 4: $3(4) + 2 = 14$ (1); Figure 5: $3(5) + 2 = 17$ (1); Difference $= 17 - 14 = 3$ (1). | 1+1+1 |
| | **OR:** $1,25,000 \times 8 = 10,00,000$ (1); Words: ten lakh (1); International: 1,000,000 (1). | 1+1+1 |
| 25 | Cloth left $= 60 - 12 \times 3$; $= 24$ m (1); Cloth used $= 36$ m (1); Cost $= 36 \times 45 = 1620$ (1). | 1+1+1 |
| | **OR:** Divide: $3 \div 8 = 0.375$ (1); Compare $0.375$ with $0.4$ (1); $\frac{3}{8}$ is less than $0.4$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Annual use $= 1,250 \times 365 = 4,56,250$ litres (1); Nearest lakh: $5,00,000$ (1); In words: five lakh (1); Check by estimation: 365 is about 400 (2). | 1+1+1+2 |
| | **OR:** Cost price $= 20 \times 80 + 15 \times 120$ (1); $= 3400$ (1); Profit/loss $= 3700 - 3400$ (1); $= 300$ (1); Profit (1). | 1+1+1+1+1 |
| 27 | Rice $= 2.5 \times 48 = 120$ (1); Wheat $= 1.5 \times 36 = 54$ (1); Total $= 174$ (1); Balance $= 200 - 174 = 26$ (2). | 1+1+1+2 |

| | | |
|---|---|---|
| | **OR:** Notebooks cost $12n$ and pens cost $20$ (1); Total cost $= 12n + 20$ (1); At $n = 10$: $140$ (1); Change $= 200 - 140 = 60$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $5x$ | Subtract the coefficients of the like terms. |
| 2 | **(d)** 5 | Count the zeros. |
| 3 | **(b)** 38 | Evaluate both. |
| 4 | **(d)** 725 | ₹1 = 100 paise. |
| 5 | **(b)** 14 | Add inside the bracket first. |
| 6 | **(b)** 99,999 | All digits are 9. |
| 7 | **(c)** $4 + 6 \times 5$ | Evaluate each option. |
| 8 | **(d)** 4.4 | Look at the hundredths digit. |
| 9 | **(d)** $4(x + 3)$ | Use brackets to keep the order of operations. |
| 10 | **(a)** 76,54,322 | Subtract. |
| 11 | **(c)** 28 | Money paid minus cost. |
| 12 | **(d)** 3.75 | Work out the bracket first. |
| 13 | **(c)** 110 | Fare = fixed charge + rate × distance. |
| 14 | **(d)** 400 | 1 lakh = 100 thousand. |
| 15 | **(b)** 29 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $7 \times 9 + 12$ (1); Value $= 75$ (1). | 1+1 |
| 17 | Make the divisor a whole number: $36 \div 4$ (1); Answer: $9$ (1). | 1+1 |
| 18 | Collect like terms: $(7 - 3)x$ and $(4 + 6)$ (1); Answer: $4x + 10$ (1). | 1+1 |
| 19 | 34,56,789 rounds to 35,00,000; 12,34,567 rounds to 12,00,000 (1); Estimate $= 47,00,000$ (1). | 1+1 |
| 20 | First $= 64$; second $= 64$ (1); Yes, equal (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Group the $x$-terms and the $y$-terms (1); $x$-terms: $5x + 3x = 8x$ (1); Sum $= 8x + 5y$ (1). | 1+1+1 |
| 22 | Increase $= 25,67,890 - 23,45,678 = 2,22,212$ (1); Estimate: $2,20,000$ (1); State the answer in words: two lakh twenty-two thousand two hundred twelve (1). | 1+1+1 |
| 23 | $16 \times 100 = 1600$ (1); $16 \times 2 = 32$ (1); $16 \times 98 = 1568$ (1). | 1+1+1 |
| 24 | Number of pieces $= 7.5 \div 0.25$ (1); $= 30$ pieces (1); Check: $30 \times 0.25 = 7.5$ (1). | 1+1+1 |
| | **OR:** Father's age $= 3s + 2$ (1); At $s = 10$: $3(10) + 2$ (1); Father's age $= 32$ years (1). | 1+1+1 |
| 25 | Break the number by place value (1); $5 \times 10^6 + 2 \times 10^4 + 5 \times 10^3$ (1); In lakhs: $50.2$ lakh (1). | 1+1+1 |
| | **OR:** Innermost bracket first; Then outward (1); Then multiply/divide, add/subtract (1); Value $= 20$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | People surveyed $= \frac{4}{100} \times 2,50,00,000$ (1); $= 10,00,000$ (2); In words: ten lakh (2). | 1+2+2 |
| | **OR:** Brackets (1); Multiplication and division from left to right (1); Addition and subtraction from left to right (1); Value $= 13$ (1); Order of operations (1). | 1+1+1+1+1 |
| 27 | Total $= 8.5$ km (1); Longest $= 3.25$ km, shortest $= 2.5$ km (1); Difference $= 0.75$ km (1); Remaining $= 10 - 8.5 = 1.5$ km (2). | 1+1+1+2 |
| | **OR:** Total $= 8r + 6$ (1); For $r = 12$: $102$ (2); For $r = 14$: $118$ (2). | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 0.375 | Divide the numerator by the denominator. |
| 2 | **(d)** 19 | Substitute the value of $n$. |
| 3 | **(a)** 700 | 1 crore = 100 lakh. |
| 4 | **(a)** 26 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 5 | **(b)** 4.2 | Multiply as whole numbers, then count the decimal places in both factors. |
| 6 | **(d)** 48 | Perimeter $= 4\times$ side. |
| 7 | **(c)** 1 | Group digits in threes. |
| 8 | **(c)** 34 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 9 | **(d)** 3 | Multiply both numbers by 10 or 100 so that the divisor becomes a whole number. |
| 10 | **(d)** $5n + 2$ | Write the rule using a letter-number. |
| 11 | **(c)** 12,345,678 | Regroup the digits in threes. |
| 12 | **(a)** $(2 + 7) \times 4 - 9$ | Check which grouping gives the value. |
| 13 | **(d)** 80 | 1 m = 100 cm. |
| 14 | **(a)** 18 | Combine the like terms first: $m + am = (1 + a)m$. |
| 15 | **(c)** 8,00,000 | Digit times its place value. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | (i) $3.62 \times 100 = 362$ cm (1); (ii) $2.45 \times 1000 = 2450$ g (1). | 1+1 |
| 17 | Substitute: $5(6) - 3(4)$ (1); $= 18$ (1). | 1+1 |
| 18 | Regroup in threes (1); 15,000,000 (1). | 1+1 |
| 19 | Evaluate inner brackets first; Then the outer operations (1); Value $= 6$ (1). | 1+1 |
| 20 | Multiply without the decimal points: $45 \times 6 = 270$ (1); Decimal places in the factors: $2 + 1 = 3$, so the product is $0.27$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Cost $= 86.4 \times 2.5$ (1); Cost $=$ ₹216 (2). | 1+2 |
| 22 | Figure 6: $3(6) + 2 = 20$ (1); Figure 7: $3(7) + 2 = 23$ (1); Difference $= 23 - 20 = 3$ (1). | 1+1+1 |
| 23 | $86,000 \times 15 = 12,90,000$ (1); Words: twelve lakh ninety thousand (1); International: 1,290,000 (1). | 1+1+1 |
| 24 | Cloth left $= 50 - 8 \times 4$; $= 18$ m (1); Cloth used $= 32$ m (1); Cost $= 32 \times 60 = 1920$ (1). | 1+1+1 |
| | **OR:** Divide: $7 \div 20 = 0.35$ (1); Compare $0.35$ with $0.3$ (1); $\frac{7}{20}$ is greater than $0.3$ (1). | 1+1+1 |
| 25 | Group the $x$-terms and the $y$-terms (1); $x$-terms: $7x + 2x = 9x$ (1); Sum $= 9x + 5y$ (1). | 1+1+1 |
| | **OR:** Increase $= 47,89,432 - 45,00,000 = 2,89,432$ (1); Estimate: $2,90,000$ (1); State the answer in words: two lakh eighty-nine thousand four hundred thirty-two (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Annual use $= 2,400 \times 365 = 8,76,000$ litres (1); Nearest lakh: $9,00,000$ (1); In words: nine lakh (1); Check by estimation: 365 is about 400 (2). | 1+1+1+2 |
| | **OR:** Cost price $= 25 \times 60 + 10 \times 150$ (1); $= 3000$ (1); Profit/loss $= 3200 - 3000$ (1); $= 200$ (1); Profit (1). | 1+1+1+1+1 |
| 27 | Rice $= 3.5 \times 52 = 182$ (1); Wheat $= 2 \times 40 = 80$ (1); Total $= 262$ (1); Balance $= 300 - 262 = 38$ (2). | 1+1+1+2 |

| | | |
|---|---|---|
| | **OR:** Notebooks cost $15n$ and pens cost $24$ (1); Total cost $= 15n + 24$ (1); At $n = 8$: $144$ (1); Change $= 200 - 144 = 56$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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